

VAJRA

Visual Audio Joint-modal Rendering Architecture

VAJRA 2.0

Document 3 of 5

Operator's User Manual

Every module, every control, and how to read the result

WHO THIS IS FOR

The operator at the interface. No programming assumed.

WHAT IT COVERS

- A section per module: what each control does
- Reading the routing and detectability report
- Recommended settings and task recipes
- Troubleshooting, and the limits worth knowing

THE FULL SET

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|--|---------------------------|
| 1. System Requirements and Sizing | specifying the machine |
| 2. Deployment and Operations Guide | installing and running it |
| ■ 3. Operator's User Manual | using it |
| 4. Technical Writeup | understanding it |
| 5. Verification and Test Documentation | maintaining it |

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Offline synthetic-media platform · for authorised research and capability development

1. Before You Start

```
bash run.sh
```

Then open **http://localhost:7860** in a browser on that machine.

Five things to know first

- **One job at a time.** GPU work is deliberately serialised. Press a second button and it waits rather than competing for memory. Nothing has hung.
- **The first run of any module is slow.** Models load on first use and some are several gigabytes. Later runs in the same session are much faster.
- **Every result is saved automatically** to outputs/ as vajra_<what>_<timestamp>. You do not need to download it to keep it.
- **Some modules may not be installed.** The platform installs selectively. A module that reports a missing environment is a setup choice, not a fault — the message names what to add.
- **Nothing leaves the machine.** There is no cloud service behind any module.

Your source material decides the result

Property	Aim for	Why
Framing	Front-facing, whole face visible	Profiles and partial faces degrade every face model
Lighting	Even, no hard shadow across the face	Shadow edges confuse detection and show as artefacts
Audio	Clean speech, little background noise	The voice reference comes from your recording; noise is cloned too
People in frame	One speaker	The lip-sync models cannot tell which mouth to drive
Length	1–3 minutes for a first trial	Longer works, but takes proportionally longer to check

2. Module 01 — Video Edit

What it does. Transcribes a talking-head video, lets you change the words, then re-voices only what you changed in the speaker's own voice and re-synchronises their lips to match.

Why it is not the same as re-recording. Everything you do not change is left exactly as filmed — audio outside an edit is untouched to the last sample, and only the frames spanning an edit are re-rendered.

Step by step

1. Upload a talking-head video.
2. Press **Extract Transcript**. This takes up to a minute or two. The transcript appears in two panes: the left is a read-only reference, the right is yours to edit.
3. Edit the right-hand pane. Change words; keep one segment per line and do not merge or delete lines.
4. Press **Apply Edits & Re-sync**.
5. Review: the video appears top right, with the original and edited audio below it for comparison.
6. Open **Routing & detectability report** to see how the edit was made and how convincing it measures. Section 9 explains it.

Main controls

Control	Choices / range	Default	What it does
Lip-sync	Best quality · Fast	Best quality	Best quality regenerates the mouth region; Fast is more robust on difficult footage, at lower quality
Diffusion steps	10 – 50	20	Refinement passes for the mouth. Higher is slower with diminishing return
Guidance	1.0 – 3.0	1.5	How strongly the audio drives the mouth. Raise slightly if sync looks loose; too high looks exaggerated

Advanced panel — realism and routing

These defaults are deliberate. Change one at a time so you can tell what it did.

Control	Range	Default	When to change it
Match the splice to its surroundings	on / off	on	Leave on. Off skips loudness, tone, room and noise matching — useful only to hear what the matching is doing
Word-level edits	on / off	on	Leave on. Off regenerates a whole line for any change, discarding audio that could have stayed genuine
Re-sync only edited windows	on / off	on	Leave on. Off re-renders every frame, far slower, and touches footage that had no reason to change
Separate background audio	Never · Auto · Always	Never	Use Auto or Always only when music or ambience must be protected. It is a lossy round trip
Max time-stretch	0.00 – 0.30	0.15	How far replacement speech may be squeezed or stretched to fit. Above about 0.15 stretching becomes audible
Quality tier	Auto · A · B · C · D	Auto (best available)	A request, not a command: a tier your language or installation cannot deliver falls back, and the report says so
Allow non-commercial engines	on / off	off	Only if your use permits a non-commercial licence

How much can you change?

The most important limit to understand. Replacement audio must occupy the same time as the words it replaces, because the soundtrack has to stay in step with the picture.

If your new wording is...	What happens	What to do
about the same length	Fits directly. Best results.	Nothing
slightly longer or shorter	Absorbs neighbouring silence, or is gently stretched within the limit	Nothing
noticeably longer	The whole line is re-voiced rather than cut short; reported in the panel	Acceptable, but more of the line becomes synthetic
much longer (say, double)	Cannot fit — you will see <i>duration capped</i>	Reword closer to the original length

Tip. Change **a few words per line** rather than whole lines wherever the meaning allows. That is where this system's quality lives: a two-word change keeps most of the real recording, and real recording never needs disguising.

3. Module 02 — Voice Generation

What it does. Audio-only work, in two sub-tabs. **Voice Edit** changes words inside an existing recording. **Voice Clone** speaks new text in a voice taken from a reference clip.

Note. This module is **English only** in this build and needs its own environment and weights. Its first start loads several models and takes a few minutes — that is normal, not a failure.

Voice Edit

1. Upload the audio to edit.
2. Press **Extract transcript**. Do not type it by hand unless you must — a transcript that does not match the audio will misplace the edit.
3. Change only the words you want re-spoken.
4. Press **Edit the speech**.

Control	Range	Default	What it does
Alignment granularity	Word (precise) · Segment (safer)	Word (precise)	How finely text is matched to audio. Use Segment if word alignment misplaces the edit
Mask expansion	0.00 – 0.50	0.10	How far beyond the changed words the model may regenerate. Higher blends the join better but keeps less of the original

Voice Clone

Control	Range	Default	What it does
Reference clip	a few seconds of clean speech	—	Quality here matters more than any setting on this tab
Emotion guidance	0.0 – 5.0	1.0	How closely the reference's expressiveness is followed. Too high sounds theatrical
Voice guidance	0.0 – 5.0	1.0	How tightly the output holds to the reference voice. Too high can sound strained

Tip. A clean six-second reference beats a noisy thirty-second one. Whatever is in the reference is copied, including room echo and background noise.

4. Module 03 — Image Generation

Control	Range	Default	Guidance
Model	Photoreal (recommended) · General purpose	Photoreal	Photoreal for people and scenes; General purpose for illustration or unusual subjects
Prompt	free text	—	Describe subject, setting, lighting and framing
Negative prompt	free text	—	Things to avoid
Width / Height	512 – 1536, step 64	1024 × 1024	Stay near the default; extreme aspect ratios distort anatomy
Steps	4 – 50	30	Below about 20 looks unfinished; above about 35 rarely helps
Guidance	0.0 – 9.0	6.0	How strictly the prompt is followed. Low drifts; high looks harsh and over-saturated

Control	Range	Default	Guidance
Seed	–1 for random, or any number	–1	Fix it to reproduce an image exactly, or to change one setting while holding the composition

Tip. To refine an image you like: note its seed, keep it fixed, and change one thing at a time. With a random seed every change alters the whole picture and you learn nothing.

5. Module 04 — Face Swap

Control	Choices	Default	What it does
Source face (copy FROM)	image	—	The identity to copy. A clear, front-facing photo
Target type	Image · Video	Image	Video processes every frame and re-attaches the original audio
Target (paste INTO)	image or video	—	What receives the face
Face enhancer	Standard · High detail · None	Standard	Restores detail the swap cannot produce. High detail is sharper but image-only and non-commercially licensed; None leaves a visibly soft face

Note. Enhancement is not cosmetic. The swap works on a small aligned crop, so its output has a resolution ceiling; without restoration the face reads soft against a sharp scene. Leave it on **Standard** unless you have a reason.

6. Module 05 — Text → Video

With a prompt alone it generates a short video with its own synchronised audio. With a reference photo it animates that photo, preserving the person's identity.

Control	Range	Default	Guidance
Reference photo	optional	none	With a photo, identity is preserved and the photo becomes the opening frame
Motion-video engine	Identity-preserving · Synchronized audio	Identity-preserving	Used only when a photo is given. Identity-preserving is better for people; the other is faster and adds a soundtrack
Resolution	360p · 480p · 576p · 720p	480p (832×480)	Raise only if you have VRAM and RAM to spare — cost climbs steeply
Duration (seconds)	1 – 4	3	Short clips by design. Longer is much slower
Frame rate	25 – 60	25	Higher is smoother but generates proportionally more frames
Steps (audio engine)	10 – 50	40	Refinement passes
Guidance (identity engine)	1.0 – 5.0	1.0	Prompt adherence for the identity engine

Caution. The heaviest module by a wide margin. Both engines are very large and rely on CPU offload, so system RAM matters as much as VRAM. If it is not installed, that was a deliberate setup choice.

7. Module 06 — Image Edit

Control	Range	Default	Guidance
Image(s) to edit	1 – 3	—	With several, refer to them by number in the instruction

Control	Range	Default	Guidance
Edit instruction	free text	—	Say what should change. Instructions work better than descriptions of the desired final state
Negative prompt	free text	—	Optional
Steps	10 – 50	40	Refinement passes
Guidance	1.0 – 8.0	4.0	How literally the instruction is followed. Too high distorts the untouched parts
Seed	–1 or a number	–1	Fix to reproduce a result

8. Module 07 — Media Studio

Preparation and conversion tools that use **no GPU at all**. Use these to get material into shape before spending GPU time on it.

Tool	Inputs	Use it to
Trim video / audio	file, start and end in seconds	Cut to the section you need — the most effective way to reduce processing time
Grab frame	video, time in seconds	Take a still for a face-swap source or a video reference photo
Clean audio	audio or video	Reduce background noise before cloning a voice from it
Change background	portrait; white, green, gray or black	Replace a portrait's background
Burn captions	video	Write subtitles permanently into the picture
Convert	any file; MP4, compressed MP4, MP3, WAV	Fix an unusable format, or shrink a file
Resize	video, width and height	Match a required output size
GIF	video; fps 5–24, width 240–720	Produce a short animated GIF
Merge	two or more files, video or audio	Join clips end to end
Session outputs	—	Collect everything produced this session as one zip

Tip. Look for **Send to** buttons. They push a result straight into another module without saving and re-uploading — a grabbed frame into Face Swap, or a trimmed clip into Video Edit.

9. Reading the Quality Report

After a Video Edit, open **Routing & detectability report**. This is the part of the system that tells you whether to trust the result.

Routing — how the edit was made

Field	Meaning
Tier	The method used. A regenerates only the changed words using the surrounding audio — best. C regenerates a pause-bounded phrase. D regenerates a whole line.
Method	Whether the edit was in-place infill or phrase regeneration
Language	The language used for synthesis
Ceiling	A better tier exists for this language but is not installed — worth reporting to whoever maintains the system
Note	Anything that limited the result, in plain words

Detectability scorecard — how convincing it is

The headline is the **seam percentile**. The system measures every genuine word boundary in your own recording, builds that distribution, and reports where your edit's join falls within it. The recording becomes its own control group.

Reading	Interpretation	Action
under 50th percentile — inaudible	Smoother than most real word boundaries in this recording	None
50th – 80th — unlikely to be noticed	Within the normal range for this speaker	None
80th – 95th — possibly audible	More abrupt than most genuine boundaries	Listen carefully; consider rewording
above 95th — likely audible	More abrupt than nearly every genuine boundary	Reword, or move the edit to a natural pause

Alongside it: **loudness** (difference from the surrounding audio — under 1 dB is imperceptible), **tone** (spectral difference — under 4 dB is good), and **floor** (whether background hiss continues across the edit).

Caution. What the score does not tell you. It measures the *join*, not the quality of the replacement. A perfectly blended splice of poor synthesis still shows PASS. The report names four things it does not measure — speaker similarity, naturalness, intelligibility and lip-sync accuracy. **Always watch and listen to the result yourself.**

Lip-sync panel

Reports how many regions of video were re-rendered and what percentage of the clip that was — the rest is your original footage, untouched. If the whole video was re-synced, it says why.

10. Recommended Settings

Task	Settings
Video Edit — first attempt	All defaults. Change two or three words in a 1–2 minute clip.
Video Edit — critical work	Diffusion steps 30; Max time-stretch 0.10; keep every realism option on; reword to match the original syllable count
Video Edit — quick check	Lip-sync <i>Fast</i> , steps 10. Confirm the wording before committing to a full render

Task	Settings
Video Edit — music or ambience present	Separate background audio → Always
Voice Clone — best fidelity	6–10 seconds of clean reference; both guidance sliders at 1.0
Image Generation — photoreal person	Photoreal; 1024×1024; steps 30; guidance 6.0; a negative prompt listing artefacts you keep seeing
Face Swap — video	Front-facing source; enhancer Standard; trim the clip first
Text → Video — first attempt	480p, 3 seconds, 25 fps. Raise only after you have a result you like

11. When Something Goes Wrong

What you see	What it means	What to do
"... is not installed — missing the environment / weights"	That module was not installed on this machine	Not a fault. Ask the administrator to add that module
"Duration capped ... off by N seconds"	Your wording needs far more time than the audio it replaces	Reword shorter
Report says a whole line was re-voiced	The change was too extensive to localise	Normal. Change fewer words per line
Voice module refuses the first request	Its services had not finished starting	Wait and retry; the first start takes a few minutes
Edited voice sounds processed	Over-correction, or a poor voice reference	Check the reference is clean; as a test turn off <i>Match the splice to its surroundings</i> and compare
Mouth looks soft or blurred	The generated region is lower resolution than the face	Expected on this line; sharper mouth rendering is next-phase work
Nothing happens after pressing a button	Another GPU job is running; yours is queued	Wait. By design.
Interface unreachable	The application is not running	Restart with <code>bash run.sh</code>

12. Limits Worth Knowing

- **Editing cannot add much speech.** Replacement audio must fit the time it replaces.
- **Voice Generation is English only** in this build. Other languages fall back to a lower tier, and the report says so.
- **One speaker per edit.** The lip-sync models cannot tell which mouth belongs to which voice.
- **Clean, front-facing footage works; difficult footage degrades.** Profiles, occlusion and poor light all reduce quality.
- **The quality score measures joins, not content.** Always check the result yourself.
- **Video generation produces short clips** — a few seconds — and is the most demanding module.

Habits that pay off

- **Trim first.** Processing time scales with length.
- **Change one setting at a time**, or you cannot attribute a difference to anything.
- **Keep a note of seeds** for images you may need again.
- **Read the report every time.** Quicker than re-listening, and it tells you which stage to adjust.

- **Keep your source material.** Every result is reproducible from the input, the settings and the seed — but not from the output.